SOC NETWORK MONITORING & INTRUSION DETECTION REPORT

(Zeek + Elastic SIEM)

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1. Incident Overview

Incident Title:
Network Traffic Monitoring and Anomaly Detection using Zeek (Simulated)

Incident Type:
Network Anomaly & Intrusion Detection

Date & Time:
20–21 January 2026 (IST)

Reported By:
SOC Analyst (Lab Environment)

Environment:
Controlled SOC Lab (Simulated Network)

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2. Affected Assets

Monitored Systems:

- Ubuntu Server (Zeek Sensor & Log Forwarder)
- Windows Virtual Machine (User Traffic Source)

Network Scope:
Internal Virtual Network (Private IP Range)

Monitoring Platform:
Elastic SIEM (Kibana)

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3. Detection Method

The activity was detected through **network traffic analysis logs** generated by **Zeek** and ingested into **Elastic SIEM** via **Filebeat**.

Detection relied on:

- Connection-level logs
- DNS activity logs
- Protocol anomaly indicators
- Intrusion detection event categorization

No signature-based IDS rules were used.

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4. Incident Description

During routine network monitoring, multiple network events were observed involving:

- Repeated TCP connections between internal systems
- DNS service discovery traffic
- Abnormal TCP session behavior

Zeek identified protocol-level irregularities and flagged them as anomalous events. These events were centrally visible in Elastic SIEM for investigation.

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5. Investigation Details

The following Zeek datasets were analyzed:

a) Network Connections (`zeek.connection`)

- Source and destination IP addresses
- TCP and UDP session metadata

- Connection states and durations

b) DNS Activity (`zeek.dns`)

- Domain name resolution requests
- Multicast DNS traffic
- Local service discovery events

c) Anomaly Detection (`zeek.weird` / Intrusion Detection)

- TCP state violations
- Improper session termination
- Data transmission before session establishment

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6. Key Findings

- Normal user-generated traffic such as web browsing and DNS lookups was observed.
- Repeated TCP sessions between internal hosts were logged.
- Zeek flagged multiple protocol anomalies, including:
    - `data_before_established`
    - `active_connection_reuse`
    - `inappropriate_FIN`
- At least one event was categorized under **intrusion_detection** in Elastic SIEM.

These findings indicate abnormal but **non-exploitative** network behavior.

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7. Analysis

The detected anomalies are consistent with:

- Misordered TCP packets
- Reused TCP sessions
- Application-level behavior triggering protocol warnings

No indicators of malware exploitation, command-and-control traffic, or data exfiltration were identified.

As the activity occurred in a controlled lab environment, it is classified as **benign anomalous traffic**.

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8. Severity Assessment

Severity Level: Low to Medium

Reasoning:

- Anomalous behavior detected at protocol level
- No confirmed compromise
- No malicious payloads observed
- No persistence or lateral movement identified

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9. Response Actions

Since this was a simulated lab environment:

- No containment actions were required
- No remediation steps were applied
- Monitoring was continued for learning and validation purposes

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10. Recommendations

If similar activity is observed in a production environment:

- Investigate repeated TCP anomalies for misconfigured applications
- Correlate network anomalies with host-based logs
- Establish baseline network behavior
- Enable alerting for repeated intrusion detection events
- Monitor DNS activity for suspicious domains

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11. Incident Status

Current Status: Closed

Closure Reason:
Simulated network activity in a controlled SOC lab environment with no security impact.

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12. Analyst Notes

This incident demonstrates the value of **network-based monitoring** using Zeek for detecting subtle protocol anomalies that may not be captured by traditional signature-based IDS solutions. Centralized visibility through Elastic SIEM enables SOC analysts to correlate traffic patterns and investigate potential threats effectively.